Chapter 10

TRANSFORMATION OF RANDOM VARIABLES AND THEIR DISTRIBUTIONS

In many statistical applications, given the probability distribution of a univariate random variable X, one would like to know the probability distribution of another univariate random variable $Y = \phi(X)$, where ϕ is some known function. For example, if we know the probability distribution of the random variable X, we would like know the distribution of $Y = \ln(X)$. For univariate random variable X, some commonly used transformed random variable Y of X are: $Y = X^2$, $Y = |X|$, $Y = \sqrt{|X|}$, $Y = \ln(X)$, $Y = \frac{X-\mu}{\sigma}$, and $Y = \left(\frac{X-\mu}{\sigma}\right)^2$. Similarly for a bivariate random variable (X, Y), some of the most common transformations of X and Y are $X + Y$, XY, $\frac{X}{Y}$, $\min\{X, Y\}$, $\max\{X, Y\}$ or $\sqrt{X^2 + Y^2}$. In this chapter, we examine various methods for finding the distribution of a transformed univariate or bivariate random variable, when transformation and distribution of the variable are known. First, we treat the univariate case. Then we treat the bivariate case.

We begin with an example for univariate discrete random variable.

Example 10.1. The probability density function of the random variable X is shown in the table below.

x	-2	-1	0	1	2	3	4
$f(x)$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$

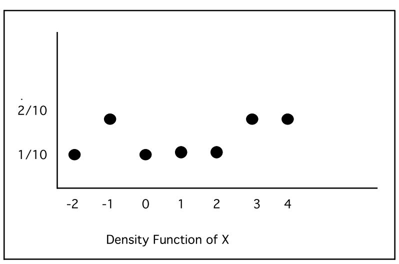
What is the probability density function of the random variable $Y = X^2$?

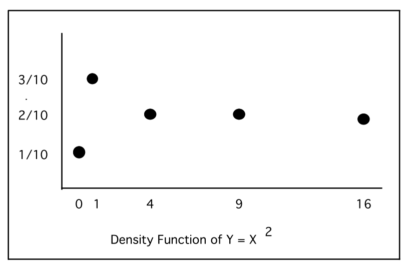
Answer: The space of the random variable X is $R_X = \{-2, -1, 0, 1, 2, 3, 4\}$. Then the space of the random variable Y is $R_Y = \{x^2 \mid x \in R_X\}$. Thus, $R_Y = \{0, 1, 4, 9, 16\}$. Now we compute the probability density function $g(y)$ for y in R_Y.

$$\begin{aligned}
g(0) &= P(Y = 0) = P(X^2 = 0) = P(X = 0)) = \frac{1}{10} \\
g(1) &= P(Y = 1) = P(X^2 = 1) = P(X = -1) + P(X = 1) = \frac{3}{10} \\
g(4) &= P(Y = 4) = P(X^2 = 4) = P(X = -2) + P(X = 2) = \frac{2}{10} \\
g(9) &= P(Y = 9) = P(X^2 = 9) = P(X = 3) = \frac{2}{10} \\
g(16) &= P(Y = 16) = P(X^2 = 16) = P(X = 4) = \frac{2}{10}.
\end{aligned}$$

We summarize the distribution of Y in the following table.

y	0	1	4	9	16
$g(y)$	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{2}{10}$





Example 10.2. The probability density function of the random variable X is shown in the table below.

x	1	2	3	4	5	6
$f(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

What is the probability density function of the random variable $Y = 2X + 1$?

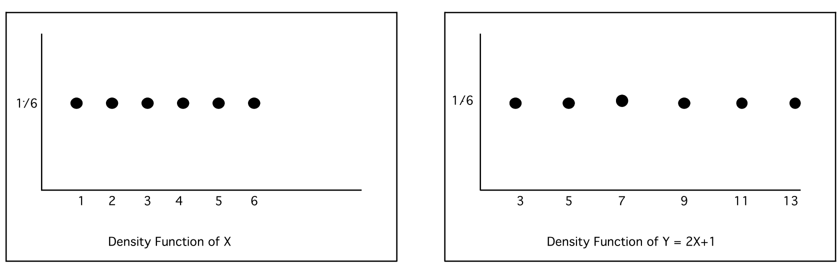
Answer: The space of the random variable X is $R_X = \{1, 2, 3, 4, 5, 6\}$. Then the space of the random variable Y is $R_Y = \{2x + 1 \mid x \in R_X\}$. Thus, $R_Y = \{3, 5, 7, 9, 11, 13\}$. Next we compute the probability density function $g(y)$ for y in R_Y. The pdf $g(y)$ is given by

$$
\begin{aligned}
g(3) &= P(Y = 3) = P(2X + 1 = 3) = P(X = 1)) = \frac{1}{6} \\
g(5) &= P(Y = 5) = P(2X + 1 = 5) = P(X = 2)) = \frac{1}{6} \\
g(7) &= P(Y = 7) = P(2X + 1 = 7) = P(X = 3)) = \frac{1}{6} \\
g(9) &= P(Y = 9) = P(2X + 1 = 9) = P(X = 4)) = \frac{1}{6} \\
g(11) &= P(Y = 11) = P(2X + 1 = 11) = P(X = 5)) = \frac{1}{6} \\
g(13) &= P(Y = 13) = P(2X + 1 = 13) = P(X = 6)) = \frac{1}{6}.
\end{aligned}
$$

We summarize the distribution of Y in the following table.

y	3	5	7	9	11	13
$g(y)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

The distribution of X and $2X + 1$ are illustrated below.



In Example 10.1, we computed the distribution (that is, the probability density function) of transformed random variable $Y = \phi(X)$, where $\phi(x) = x^2$. This transformation is not either increasing or decreasing (that is, monotonic) in the space, R_X, of the random variable X. Therefore, the distribution of Y turn out to be quite different from that of X. In Example 10.2, the form of distribution of the transform random variable $Y = \phi(X)$, where $\phi(x) = 2x + 1$, is essentially same. This is mainly due to the fact that $\phi(x) = 2x + 1$ is monotonic in R_X.

In this chapter, we shall examine the probability density function of transformed random variables by knowing the density functions of the original random variables. There are several methods for finding the probability density function of a transformed random variable. Some of these methods are:

(1) distribution function method
(2) transformation method
(3) convolution method, and
(4) moment generating function method.

Among these four methods, the transformation method is the most useful one. The convolution method is a special case of this method. The transformation method is derived using the distribution function method.

10.1. Distribution Function Method

We have seen in chapter six that an easy way to find the probability density function of a transformation of continuous random variables is to determine its distribution function and then its density function by differentiation.

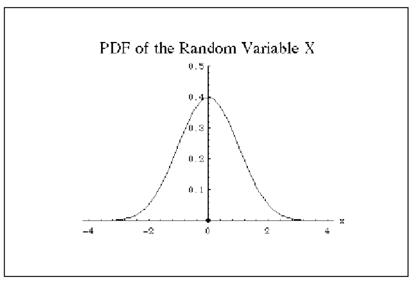
Example 10.3. A box is to be constructed so that the height is 4 inches and its base is X inches by X inches. If X has a standard normal distribution, what is the distribution of the volume of the box?

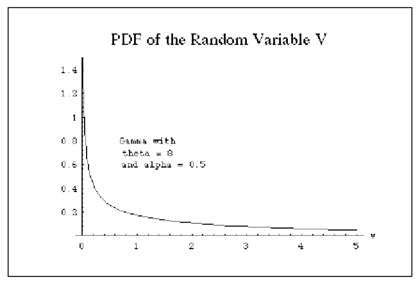
Answer: The volume of the box is a random variable, since X is a random variable. This random variable V is given by $V = 4X^2$. To find the density function of V, we first determine the form of the distribution function $G(v)$ of V and then we differentiate $G(v)$ to find the density function of V. The distribution function of V is given by

$$
\begin{aligned}
G(v) &= P\,(V \leq v) \\
&= P\left(4X^2 \leq v\right) \\
&= P\left(-\frac{1}{2}\sqrt{v} \leq X \leq \frac{1}{2}\sqrt{v}\right) \\
&= \int_{-\frac{1}{2}\sqrt{v}}^{\frac{1}{2}\sqrt{v}} \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}x^2}\, dx \\
&= 2\int_0^{\frac{1}{2}\sqrt{v}} \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}x^2}\, dx \qquad \text{(since the integrand is even).}
\end{aligned}
$$

Hence, by the Fundamental Theorem of Calculus, we get

$$
\begin{aligned}
g(v) &= \frac{dG(v)}{dv} \\
&= \frac{d}{dv}\left(2 \int_0^{\frac{1}{2}\sqrt{v}} \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}x^2}\, dx \right) \\
&= 2\, \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{1}{2}\sqrt{v}\right)^2} \left(\frac{1}{2}\right) \frac{d\sqrt{v}}{dv} \\
&= \frac{1}{\sqrt{2\pi}}\; e^{-\frac{1}{8}v}\; \frac{1}{2\sqrt{v}} \\
&= \frac{1}{\Gamma\left(\frac{1}{2}\right)\, \sqrt{8}}\; v^{\frac{1}{2}-1}\; e^{-\frac{v}{8}} \\
&= V \sim GAM\left(8,\, \frac{1}{2}\right).
\end{aligned}
$$





Example 10.4. If the density function of X is

$$
f(x) = \begin{cases} \frac{1}{2} & \text{for } -1 < x < 1 \\ \\ 0 & \text{otherwise,} \end{cases}
$$

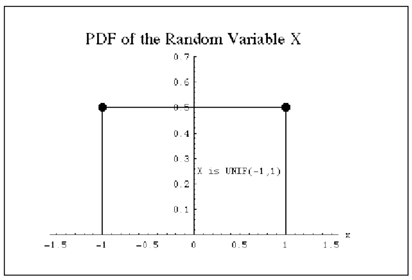
what is the probability density function of $Y = X^2$?

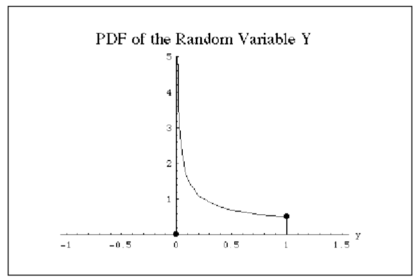
Answer: We first find the cumulative distribution function of Y and then by differentiation, we obtain the density of Y. The distribution function $G(y)$ of Y is given by

$$
\begin{aligned}
G(y) &= P\left(Y \leq y\right) \\
&= P\left(X^2 \leq y\right) \\
&= P\left(-\sqrt{y} \leq X \leq \sqrt{y}\right) \\
&= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2}\, dx \\
&= \sqrt{y}.
\end{aligned}
$$

Hence, the density function of Y is given by

$$\begin{aligned} g(y) &= \frac{dG(y)}{dy} \\ &= \frac{d\sqrt{y}}{dy} \\ &= \frac{1}{2\sqrt{y}} \qquad\qquad \text{for} \quad 0 < y < 1. \end{aligned}$$





10.2. Transformation Method for Univariate Case

The following theorem is the backbone of the transformation method.

Theorem 10.1. Let X be a continuous random variable with probability density function $f(x)$. Let $y = T(x)$ be an increasing (or decreasing) function. Then the density function of the random variable $Y = T(X)$ is given by

$$g(y) = \left| \frac{dx}{dy} \right| f(W(y))$$

where $x = W(y)$ is the inverse function of $T(x)$.

Proof: Suppose $y = T(x)$ is an increasing function. The distribution function $G(y)$ of Y is given by

$$\begin{aligned} G(y) &= P\left(Y \le y\right) \\ &= P\left(T(X) \le y\right) \\ &= P\left(X \le W(y)\right) \\ &= \int_{-\infty}^{W(y)} f(x)\, dx. \end{aligned}$$

Then, differentiating we get the density function of Y, which is

$$\begin{aligned} g(y) &= \frac{dG(y)}{dy} \\ &= \frac{d}{dy}\left(\int_{-\infty}^{W(y)} f(x)\,dx\right) \\ &= f(W(y))\,\frac{dW(y)}{dy} \\ &= f(W(y))\,\frac{dx}{dy} \qquad\qquad (\text{since} \quad x = W(y)). \end{aligned}$$

On the other hand, if $y = T(x)$ is a decreasing function, then the distribution function of Y is given by

$$\begin{aligned} G(y) &= P\left(Y \le y\right) \\ &= P\left(T(X) \le y\right) \\ &= P\left(X \ge W(y)\right) \qquad (\text{since } T(x) \text{ is decreasing}) \\ &= 1 - P\left(X \le W(y)\right) \\ &= 1 - \int_{-\infty}^{W(y)} f(x)\,dx. \end{aligned}$$

As before, differentiating we get the density function of Y, which is

$$\begin{aligned} g(y) &= \frac{dG(y)}{dy} \\ &= \frac{d}{dy}\left(1 - \int_{-\infty}^{W(y)} f(x)\,dx\right) \\ &= -f(W(y))\,\frac{dW(y)}{dy} \\ &= -f(W(y))\,\frac{dx}{dy} \qquad\qquad (\text{since} \quad x = W(y)). \end{aligned}$$

Hence, combining both the cases, we get

$$g(y) = \left|\frac{dx}{dy}\right|\, f(W(y))$$

and the proof of the theorem is now complete.

Example 10.5. Let $Z = \frac{X-\mu}{\sigma}$. If $X \sim N\left(\mu,\, \sigma^2\right)$, what is the probability density function of Z?

Answer:

$$z = U(x) = \frac{x-\mu}{\sigma}.$$

Hence, the inverse of U is given by

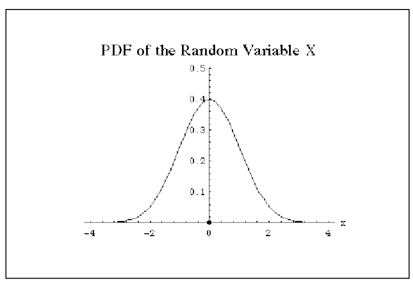
$$\begin{aligned} W(z) &= x \\ &= \sigma\, z + \mu. \end{aligned}$$

Therefore

$$\frac{dx}{dz} = \sigma.$$

Hence, by Theorem 10.1, the density of Z is given by

$$\begin{aligned} g(z) &= \left|\frac{dx}{dz}\right| \; f(W(y)) \\ &= \sigma \, \frac{1}{\sqrt{2\pi\sigma^2}} \, e^{-\frac{1}{2}\left(\frac{W(z)-\mu}{\sigma}\right)^2} \\ &= \frac{1}{\sqrt{2\pi}} \; e^{-\frac{1}{2}\left(\frac{z\sigma+\mu-\mu}{\sigma}\right)^2} \\ &= \frac{1}{\sqrt{2\pi}}\, e^{-\frac{1}{2}\, z^2}. \end{aligned}$$



Example 10.6. Let $Z = \frac{X-\mu}{\sigma}$. If $X \sim N\left(\mu, \sigma^2\right)$, then show that Z^2 is chi-square with one degree of freedom, that $Z^2 \sim \chi^2(1)$.

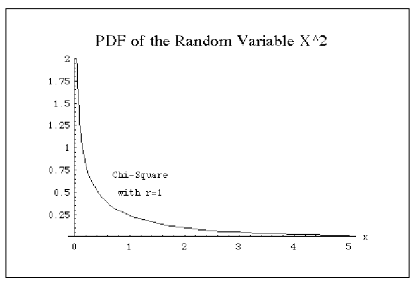
Answer:

$$\begin{aligned} y = T(x) &= \left(\frac{x-\mu}{\sigma}\right)^2. \\ x &= \mu + \sigma\sqrt{y}. \\ W(y) &= \mu + \sigma\,\sqrt{y}, \qquad\qquad y > 0. \\ \frac{dx}{dy} &= \frac{\sigma}{2\sqrt{y}}. \end{aligned}$$

The density of Y is

$$
\begin{aligned}
g(y) &= \left|\frac{dx}{dy}\right| \, f(W(y)) \\
&= \sigma \, \frac{1}{2\,\sqrt{y}} \, f(W(y)) \\
&= \sigma \, \frac{1}{2\,\sqrt{y}} \, \frac{1}{\sqrt{2\pi\sigma^2}} \, e^{-\frac{1}{2}\left(\frac{W(y)-\mu}{\sigma}\right)^2} \\
&= \frac{1}{2\sqrt{2\,\pi\,y}} \, e^{-\frac{1}{2}\left(\frac{\sqrt{y}\sigma+\mu-\mu}{\sigma}\right)^2} \\
&= \frac{1}{2\sqrt{2\,\pi\,y}} \, e^{-\frac{1}{2}y} \\
&= \frac{1}{2\sqrt{\pi}\,\sqrt{2}} \, y^{-\frac{1}{2}} \, e^{-\frac{1}{2}y} \\
&= \frac{1}{2\Gamma\left(\frac{1}{2}\right)\,\sqrt{2}} \, y^{-\frac{1}{2}} \, e^{-\frac{1}{2}y}.
\end{aligned}
$$

Hence $Y \sim \chi^2(1)$.



Example 10.7. Let $Y = -\ln X$. If $X \sim UNIF(0,1)$, then what is the density function of Y where nonzero?

Answer: We are given that

$$y = T(x) = -\ln x.$$

Hence, the inverse of $y = T(x)$ is given by

$$
\begin{aligned}
W(y) &= x \\
&= e^{-y}.
\end{aligned}
$$

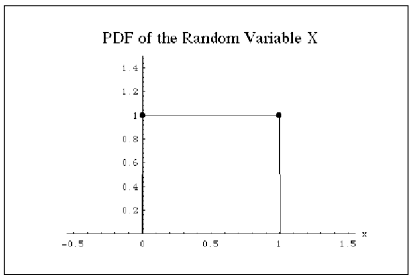
Therefore

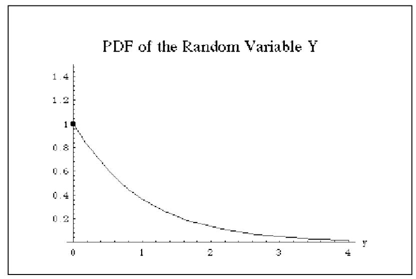
$$\frac{dx}{dy} = -\,e^{-y}.$$

Hence, by Theorem 10.1, the probability density of Y is given by

$$\begin{aligned} g(y) &= \left|\frac{dx}{dy}\right| f(W(y)) \\ &= e^{-y} f(W(y)) \\ &= e^{-y}. \end{aligned}$$

Thus $Y \sim EXP(1)$. Hence, if $X \sim UNIF(0,1)$, then the random variable $-\ln X \sim EXP(1)$.





Although all the examples we have in this section involve continuous random variables, the transformation method also works for the discrete random variables.

10.3. Transformation Method for Bivariate Case

In this section, we extend the Theorem 10.2 to the bivariate case and present some examples to illustrate the importance of this extension. We state this theorem without a proof.

Theorem 10.2. Let X and Y be two continuous random variables with joint density $f(x,y)$. Let $U = P(X,Y)$ and $V = Q(X,Y)$ be functions of X and Y. If the functions $P(x,y)$ and $Q(x,y)$ have single valued inverses, say $X = R(U,V)$ and $Y = S(U,V)$, then the joint density $g(u,v)$ of U and V is given by

$$g(u,v) = |J| \;\; f(R(u,v),\, S(u,v)),$$

where J denotes the Jacobian and given by

$$\begin{aligned} J &= \det \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} \\ &= \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}. \end{aligned}$$

Example 10.8. Let X and Y have the joint probability density function

$$f(x,y) = \begin{cases} 8\,xy & \text{for } 0 < x < y < 1 \\ \\ 0 & \text{otherwise.} \end{cases}$$

What is the joint density of $U = \frac{X}{Y}$ and $V = Y$?

Answer: Since

$$\left. \begin{aligned} U &= \frac{X}{Y} \\ V &= Y \end{aligned} \right\}$$

we get by solving for X and Y

$$\left. \begin{aligned} X &= U\,Y = U\,V \\ Y &= V. \end{aligned} \right\}$$

Hence, the Jacobian of the transformation is given by

$$\begin{aligned} J &= \frac{\partial x}{\partial u}\,\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\,\frac{\partial y}{\partial u} \\ &= v \cdot 1 - u \cdot 0 \\ &= v. \end{aligned}$$

The joint density function of U and V is

$$\begin{aligned} g(u,v) &= |J|\ f(R(u,v),\, S(u,v)) \\ &= |v|\, f(uv, v) \\ &= v\, 8\,(uv)\, v \\ &= 8\, uv^3. \end{aligned}$$

Note that, since

$$0 < x < y < 1$$

we have

$$0 < uv < v < 1.$$

The last inequalities yield

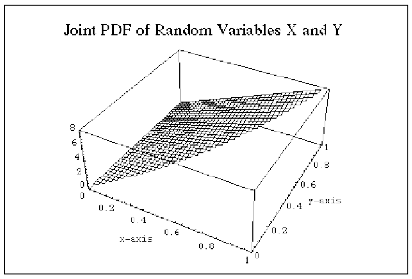
$$\left. \begin{aligned} &0 < uv < v \\ &0 < v < 1. \end{aligned} \right\}$$

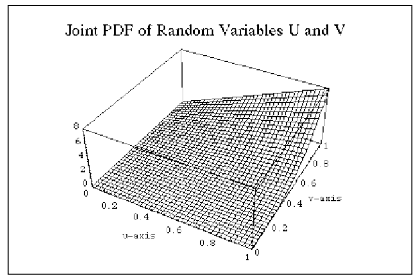
Therefore, we get

$$\left.\begin{aligned} &0 < u < 1 \\ &0 < v < 1. \end{aligned}\right\}$$

Thus, the joint density of U and V is given by

$$g(u,v) = \begin{cases} 8\,uv^3 & \text{for } 0 < u < 1;\ 0 < v < 1 \\ \\ 0 & \text{otherwise.} \end{cases}$$





Example 10.9. Let each of the independent random variables X and Y have the density function

$$f(x) = \begin{cases} e^{-x} & \text{for } 0 < x < \infty \\ \\ 0 & \text{otherwise.} \end{cases}$$

What is the joint density of $U = X$ and $V = 2X + 3Y$ and the domain on which this density is positive?

Answer: Since

$$\left.\begin{aligned} U &= X \\ V &= 2X + 3Y, \end{aligned}\right\}$$

we get by solving for X and Y

$$\left.\begin{aligned} X &= U \\ Y &= \frac{1}{3}\,V - \frac{2}{3}\,U. \end{aligned}\right\}$$

Hence, the Jacobian of the transformation is given by

$$\begin{aligned} J &= \frac{\partial x}{\partial u}\,\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\,\frac{\partial y}{\partial u} \\ &= 1 \cdot \left(\frac{1}{3}\right) - 0 \cdot \left(-\frac{2}{3}\right) \\ &= \frac{1}{3}. \end{aligned}$$

The joint density function of U and V is

$$\begin{aligned} g(u,v) &= |J| \; f(R(u,v),\, S(u,v)) \\ &= \left|\frac{1}{3}\right| \; f\left(u,\; \frac{1}{3}v - \frac{2}{3}u\right) \\ &= \frac{1}{3}e^{-u} \; e^{-\frac{1}{3}v + \frac{2}{3}u} \\ &= \frac{1}{3}\; e^{-\left(\frac{u+v}{3}\right)}. \end{aligned}$$

Since

$$0 < x < \infty$$
$$0 < y < \infty,$$

we get

$$0 < u < \infty$$
$$0 < v < \infty,$$

Further, since $v = 2u + 3y$ and $3y > 0$, we have

$$v > 2u.$$

Hence, the domain of $g(u,v)$ where nonzero is given by

$$0 < 2u < v < \infty.$$

The joint density $g(u,v)$ of the random variables U and V is given by

$$g(u,v) = \begin{cases} \frac{1}{3}\; e^{-\left(\frac{u+v}{3}\right)} & \text{for } 0 < 2u < v < \infty \\ \\ 0 & \text{otherwise.} \end{cases}$$

Example 10.10. Let X and Y be independent random variables, each with density function

$$f(x) = \begin{cases} \lambda\, e^{-\lambda x} & \text{for } 0 < x < \infty \\ \\ 0 & \text{otherwise,} \end{cases}$$

where $\lambda > 0$. Let $U = X + 2Y$ and $V = 2X + Y$. What is the joint density of U and V?

Answer: Since

$$\left.\begin{aligned} U &= X + 2Y \\ V &= 2X + Y, \end{aligned}\right\}$$

we get by solving for X and Y

$$\left.\begin{aligned} X &= -\frac{1}{3}\,U + \frac{2}{3}\,V \\ Y &= \frac{2}{3}\,U - \frac{1}{3}\,V. \end{aligned}\right\}$$

Hence, the Jacobian of the transformation is given by

$$\begin{aligned} J &= \frac{\partial x}{\partial u}\,\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\,\frac{\partial y}{\partial u} \\ &= \left(-\frac{1}{3}\right)\,\left(-\frac{1}{3}\right) - \left(\frac{2}{3}\right)\,\left(\frac{2}{3}\right) \\ &= \frac{1}{9} - \frac{4}{9} \\ &= -\frac{1}{3}. \end{aligned}$$

The joint density function of U and V is

$$\begin{aligned} g(u,v) &= |J|\; f(R(u,v),\, S(u,v)) \\ &= \left|-\frac{1}{3}\right|\; f(R(u,v))\; f(S(u,v)) \\ &= \frac{1}{3}\,\lambda\, e^{\lambda R(u,v)}\;\lambda\, e^{\lambda S(u,v)} \\ &= \frac{1}{3}\,\lambda^2\, e^{\lambda[R(u,v)+S(u,v)]} \\ &= \frac{1}{3}\,\lambda^2\, e^{-\lambda\left(\frac{u+v}{3}\right)}. \end{aligned}$$

Hence, the joint density $g(u,v)$ of the random variables U and V is given by

$$g(u,v) = \begin{cases} \frac{1}{3}\,\lambda^2\, e^{-\lambda\left(\frac{u+v}{3}\right)} & \text{for } 0 < u < \infty;\; 0 < v < \infty \\ \\ 0 & \text{otherwise.} \end{cases}$$

Example 10.11. Let X and Y be independent random variables, each with density function

$$f(x) = \frac{1}{\sqrt{2\pi}}\; e^{-\frac{1}{2}x^2}, \qquad -\infty < x < \infty.$$

Let $U = \frac{X}{Y}$ and $V = Y$. What is the joint density of U and V? Also, what is the density of U?

Answer: Since

$$\left.\begin{aligned} U &= \frac{X}{Y} \\ V &= Y, \end{aligned}\right\}$$

we get by solving for X and Y

$$\left.\begin{aligned} X &= UV \\ Y &= V. \end{aligned}\right\}$$

Hence, the Jacobian of the transformation is given by

$$\begin{aligned} J &= \frac{\partial x}{\partial u}\,\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\,\frac{\partial y}{\partial u} \\ &= v \cdot (1) - u \cdot (0) \\ &= v. \end{aligned}$$

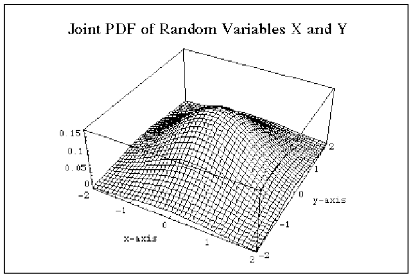
The joint density function of U and V is

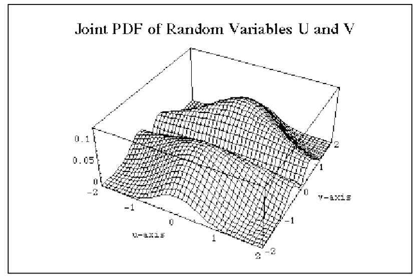
$$\begin{aligned} g(u,v) &= |J| \; f(R(u,v),\, S(u,v)) \\ &= |v| \; f(R(u,v)) \; f(S(u,v)) \\ &= |v| \; \frac{1}{\sqrt{2\pi}} \; e^{-\frac{1}{2}R^2(u,v)} \, \frac{1}{\sqrt{2\pi}} \; e^{-\frac{1}{2}S^2(u,v)} \\ &= |v| \; \frac{1}{2\pi} \, e^{-\frac{1}{2}\left[R^2(u,v)+S^2(u,v)\right]} \\ &= |v| \; \frac{1}{2\pi} \, e^{-\frac{1}{2}\left[u^2v^2+v^2\right]} \\ &= |v| \; \frac{1}{2\pi} \, e^{-\frac{1}{2}v^2\left(u^2+1\right)}. \end{aligned}$$

Hence, the joint density $g(u,v)$ of the random variables U and V is given by

$$g(u,v) = |v| \; \frac{1}{2\pi} \, e^{-\frac{1}{2}v^2\left(u^2+1\right)},$$

where $-\infty < u < \infty$ and $-\infty < v < \infty$.





Next, we want to find the density of U. We can obtain this by finding the marginal of U from the joint density of U and V. Hence, the marginal $g_1(u)$ of U is given by

$$
\begin{aligned}
g_1(u) &= \int_{-\infty}^{\infty} g(u,v)\,dv \\
&= \int_{-\infty}^{\infty} |v|\,\frac{1}{2\pi}\,e^{-\frac{1}{2}v^2\left(u^2+1\right)}\,dv \\
&= \int_{-\infty}^{0} -v\,\frac{1}{2\pi}\,e^{-\frac{1}{2}v^2\left(u^2+1\right)}\,dv + \int_{0}^{\infty} v\,\frac{1}{2\pi}\,e^{-\frac{1}{2}v^2\left(u^2+1\right)}\,dv \\
&= \frac{1}{2\pi}\,\left(\frac{1}{2}\right)\left[\frac{2}{u^2+1}\,e^{-\frac{1}{2}\,v^2\left(u^2+1\right)}\right]_{-\infty}^{0} \\
&\quad + \frac{1}{2\pi}\,\left(\frac{1}{2}\right)\left[\frac{-2}{u^2+1}\,e^{-\frac{1}{2}\,v^2\left(u^2+1\right)}\right]_{0}^{\infty} \\
&= \frac{1}{2\pi}\,\frac{1}{u^2+1} + \frac{1}{2\pi}\,\frac{1}{u^2+1} \\
&= \frac{1}{\pi\left(u^2+1\right)}.
\end{aligned}
$$

Thus $U \sim CAU(1)$.

Remark 10.1. If X and Y are independent and standard normal random variables, then the quotient $\frac{X}{Y}$ is always a Cauchy random variable. However, the converse of this is not true. For example, if X and Y are independent and each have the same density function

$$
f(x) = \frac{\sqrt{2}}{\pi}\,\frac{x^2}{1+x^4}, \qquad\qquad -\infty < x < \infty,
$$

then it can be shown that the random variable $\frac{X}{Y}$ is a Cauchy random variable. Laha (1959) and Kotlarski (1960) have given a complete description of the family of all probability density function f such that the quotient $\frac{X}{Y}$

follows the standard Cauchy distribution whenever X and Y are independent and identically distributed random variables with common density f.

Example 10.12. Let X have a Poisson distribution with mean λ. Find a transformation $T(x)$ so that $Var\,(\,T(X)\,)$ is free of λ, for large values of λ.

Answer: We expand the function $T(x)$ by Taylor's series about λ. Then, neglecting the higher orders terms for large values of λ, we get

$$T(x) = T(\lambda) + (x-\lambda)\,T'(\lambda) + \cdots\cdots$$

where $T'(\lambda)$ represents derivative of $T(x)$ at $x = \lambda$. Now, we compute the variance of $T(X)$.

$$\begin{aligned}
Var\,(\,T(X)\,) &= Var\,(\,T(\lambda) + (X-\lambda)\,T'(\lambda) + \cdots\,) \\
&= Var\,(\,T(\lambda)\,) + Var\,(\,(X-\lambda)\,T'(\lambda)\,) \\
&= 0 + \left[T'(\lambda)\right]^2\, Var(X-\lambda) \\
&= \left[T'(\lambda)\right]^2\, Var(X) \\
&= \left[T'(\lambda)\right]^2\, \lambda.
\end{aligned}$$

We want $Var(\,T(X)\,)$ to be free of λ for large λ. Therefore, we have

$$\left[T'(\lambda)\right]^2\, \lambda = k,$$

where k is a constant. From this, we get

$$T'(\lambda) = \frac{c}{\sqrt{\lambda}},$$

where $c = \sqrt{k}$. Solving this differential equation, we get

$$\begin{aligned}
T(\lambda) &= c\int \frac{1}{\sqrt{\lambda}}\,d\lambda \\
&= 2c\,\sqrt{\lambda}.
\end{aligned}$$

Hence, the transformation $T(x) = 2c\sqrt{x}$ will free $Var(\,T(X)\,)$ of λ if the random variable $X \sim POI(\lambda)$.

Example 10.13. Let $X \sim POI(\lambda_1)$ and $Y \sim POI(\lambda_2)$. What is the probability density function of $X+Y$ if X and Y are independent?

Answer: Let us denote $U = X+Y$ and $V = X$. First of all, we find the joint density of U and V and then summing the joint density we determine the marginal of U which is the density function of $X+Y$? Now writing X and Y in terms of U and V, we get

$$\left.\begin{aligned} X &= V \\ Y &= U - X = U - V. \end{aligned}\right\}$$

Hence, the Jacobian of the transformation is given by

$$\begin{aligned} J &= \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} \\ &= (0)(-1) - (1)(1) \\ &= -1. \end{aligned}$$

The joint density function of U and V is

$$\begin{aligned} g(u,v) &= |J|\; f(R(u,v),\, S(u,v)) \\ &= |-1|\; f(v, u-v) \\ &= f(v)\, f(u-v) \\ &= \left(\frac{e^{-\lambda_1}\lambda_1^v}{v!}\right)\left(\frac{e^{-\lambda_2}\lambda_2^{u-v}}{(u-v)!}\right) \\ &= \frac{e^{-(\lambda_1+\lambda_2)}\lambda_1^v\lambda_2^{u-v}}{(v)!\,(u-v)!}, \end{aligned}$$

where $v = 0, 1, 2, ..., u$ and $u = 0, 1, 2, ..., \infty$. Hence, the marginal density of U is given by

$$\begin{aligned} g_1(u) &= \sum_{v=0}^{u} \frac{e^{-(\lambda_1+\lambda_2)}\lambda_1^v\lambda_2^{u-v}}{(v)!\,(u-v)!} \\ &= e^{-(\lambda_1+\lambda_2)} \sum_{v=0}^{u} \frac{\lambda_1^v\lambda_2^{u-v}}{(v)!\,(u-v)!} \\ &= e^{-(\lambda_1+\lambda_2)} \sum_{v=0}^{u} \frac{1}{u!}\binom{u}{v}\lambda_1^v\lambda_2^{u-v} \\ &= \frac{e^{-(\lambda_1+\lambda_2)}}{u!}\,(\lambda_1+\lambda_2)^u\,. \end{aligned}$$

Thus, the density function of $U = X+Y$ is given by

$$g_1(u) = \begin{cases} \frac{e^{-(\lambda_1+\lambda_2)}}{u!}\,(\lambda_1+\lambda_2)^u & \text{for } u = 0, 1, 2, ..., \infty \\ 0 & \text{otherwise.} \end{cases}$$

This example tells us that if $X \sim POI(\lambda_1)$ and $Y \sim POI(\lambda_2)$ and they are independent, then $X+Y \sim POI(\lambda_1+\lambda_2)$.

Theorem 10.3. Let the joint density of the random variables X and Y be $f(x, y)$. Then probability density functions of $X + Y$, XY, and $\frac{Y}{X}$ are given by

$$\begin{aligned} h_{X+Y}(v) &= \int_{-\infty}^{\infty} f(u, v-u)\, du \\ h_{XY}(v) &= \int_{-\infty}^{\infty} \frac{1}{|u|}\, f\left(u, \frac{v}{u}\right)\, du \\ h_{\frac{X}{Y}}(v) &= \int_{-\infty}^{\infty} |u|\, f\,(u, vu)\, du, \end{aligned}$$

respectively.

Proof: Let $U = X$ and $V = X + Y$. So that $X = R(U, V) = U$, and $Y = S(U, V) = V - U$. Hence, the Jacobian of the transformation is given by

$$J = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u} = 1.$$

The joint density function of U and V is

$$\begin{aligned} g(u, v) &= |J|\; f(R(u, v),\, S(u, v)) \\ &= f(R(u, v),\, S(u, v)) \\ &= f(u, v-u). \end{aligned}$$

Hence, the marginal density of $V = X + Y$ is given by

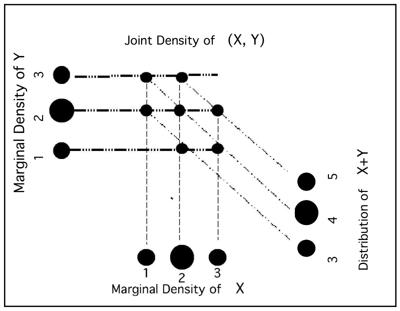
$$h_{X+Y}(v) = \int_{-\infty}^{\infty} f(u, v-u)\, du.$$

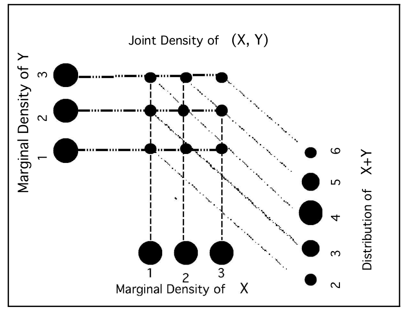
Similarly, one can obtain the other two density functions. This completes the proof.

In addition, if the random variables X and Y in Theorem 10.3 are independent and have the probability density functions $f(x)$ and $g(y)$ respectively, then we have

$$\begin{aligned} h_{X+Y}(z) &= \int_{-\infty}^{\infty} g(y)\, f(z-y)\, dy \\ h_{XY}(z) &= \int_{-\infty}^{\infty} \frac{1}{|y|}\, g(y)\, f\left(\frac{z}{y}\right)\, dy \\ h_{\frac{X}{Y}}(z) &= \int_{-\infty}^{\infty} |y|\, g(y)\, f\,(zy)\, dy. \end{aligned}$$

Each of the following figures shows how the distribution of the random variable $X + Y$ is obtained from the joint distribution of (X, Y).





Example 10.14. Roll an unbiased die twice. If X denotes the outcome in the first roll and Y denotes the outcome in the second roll, what is the distribution of the random variable $Z = \max\{X, Y\}$?

Answer: The space of X is $R_X = \{1, 2, 3, 4, 5, 6\}$. Similarly, the space of Y is $R_Y = \{1, 2, 3, 4, 5, 6\}$. Hence the space of the random variable (X, Y) is $R_X \times R_Y$. The following table shows the distribution of (X, Y).

1	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
2	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
3	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
4	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
5	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
6	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$	$\frac{1}{36}$
	1	2	3	4	5	6

The space of the random variable $Z = \max\{X, Y\}$ is $R_Z = \{1, 2, 3, 4, 5, 6\}$. Thus $Z = 1$ only if $(X, Y) = (1, 1)$. Hence $P(Z = 1) = \frac{1}{36}$. Similarly, $Z = 2$ only if $(X, Y) = (1, 2), (2, 2)$ or $(2, 1)$. Hence, $P(Z = 2) = \frac{3}{36}$. Proceeding in a similar manner, we get the distribution of Z which is summarized in the table below.

z	1	2	3	4	5	6
$h(z)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$

In this example, the random variable Z may be described as *the best out of two rolls*. Note that the probability density of Z can also be stated as

$$h(z) = \frac{2z-1}{36}, \qquad \text{for } z \in \{1,2,3,4,5,6\}.$$

10.4. Convolution Method for Sums of Random Variables

In this section, we illustrate how convolution technique can be used in finding the distribution of the sum of random variables when they are independent. This convolution technique does not work if the random variables are not independent.

Definition 10.1. Let f and g be two real valued functions. The convolution of f and g, denoted by $f \star g$, is defined as

$$\begin{aligned}(f \star g)(z) &= \int_{-\infty}^{\infty} f(z-y)\, g(y)\, dy \\ &= \int_{-\infty}^{\infty} g(z-x)\, f(x)\, dx.\end{aligned}$$

Hence from this definition it is clear that $f \star g = g \star f$.

Let X and Y be two independent random variables with probability density functions $f(x)$ and $g(y)$. Then by Theorem 10.3, we get

$$h(z) = \int_{-\infty}^{\infty} f(z-y)\, g(y)\, dy.$$

Thus, this result shows that the density of the random variable $Z = X + Y$ is the convolution of the density of X with the density of Y.

Example 10.15. What is the probability density of the sum of two independent random variables, each of which is uniformly distributed over the interval [0, 1]?

Answer: Let $Z = X + Y$, where $X \sim UNIF(0,1)$ and $Y \sim UNIF(0,1)$. Hence, the density function $f(x)$ of the random variable X is given by

$$f(x) = \begin{cases} 1 & \text{for } 0 \leq x \leq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Similarly, the density function $g(y)$ of Y is given by

$$g(y) = \begin{cases} 1 & \text{for } 0 \leq y \leq 1 \\ \\ 0 & \text{otherwise.} \end{cases}$$

Since X and Y are independent, the density function of Z can be obtained by the method of convolution. Since, the sum $z = x + y$ is between 0 and 2, we consider two cases. First, suppose $0 \leq z \leq 1$, then

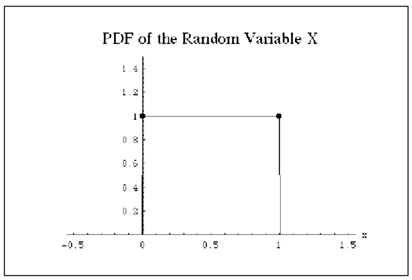
$$\begin{aligned} h(z) &= (f \star g)\,(z) \\ &= \int_{-\infty}^{\infty} f(z-x)\, g(x)\, dx \\ &= \int_{0}^{1} f(z-x)\, g(x)\, dx \\ &= \int_{o}^{z} f(z-x)\, g(x)\, dx + \int_{z}^{1} f(z-x)\, g(x)\, dx \\ &= \int_{o}^{z} f(z-x)\, g(x)\, dx + 0 \qquad \text{(since } f(z-x) = 0 \text{ between } z \text{ and } 1) \\ &= \int_{0}^{z} dx \\ &= z. \end{aligned}$$

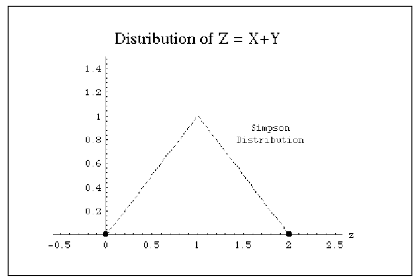
Similarly, if $1 \leq z \leq 2$, then

$$\begin{aligned} h(z) &= (f \star g)\,(z) \\ &= \int_{-\infty}^{\infty} f(z-x)\, g(x)\, dx \\ &= \int_{0}^{1} f(z-x)\, g(x)\, dx \\ &= \int_{0}^{z-1} f(z-x)\, g(x)\, dx + \int_{z-1}^{1} f(z-x)\, g(x)\, dx \\ &= 0 + \int_{z-1}^{1} f(z-x)\, g(x)\, dx \text{ (since } f(z-x) = 0 \text{ between } 0 \text{ and } z-1) \\ &= \int_{z-1}^{1} dx \\ &= 2 - z. \end{aligned}$$

Thus, the density function of $Z = X + Y$ is given by

$$h(z) = \begin{cases} 0 & \text{for } -\infty < z \leq 0 \\ z & \text{for } 0 \leq z \leq 1 \\ 2 - z & \text{for } 1 \leq z \leq 2 \\ 0 & \text{for } 2 < z < \infty \, . \end{cases}$$





The graph of this density function looks like a tent and it is called a tent function. However, in literature, this density function is known as the Simpson's distribution.

Example 10.16. What is the probability density of the sum of two independent random variables, each of which is gamma with parameter $\alpha = 1$ and $\theta = 1$?

Answer: Let $Z = X + Y$, where $X \sim GAM(1, 1)$ and $Y \sim GAM(1, 1)$. Hence, the density function $f(x)$ of the random variable X is given by

$$f(x) = \begin{cases} e^{-x} & \text{for } 0 < x < \infty \\ 0 & \text{otherwise.} \end{cases}$$

Similarly, the density function $g(y)$ of Y is given by

$$g(y) = \begin{cases} e^{-y} & \text{for } 0 < y < \infty \\ 0 & \text{otherwise.} \end{cases}$$

Since X and Y are independent, the density function of Z can be obtained by the method of convolution. Notice that the sum $z = x + y$ is between 0 and ∞, and $0 < x < z$. Hence, the density function of Z is given by

$$\begin{aligned}
h(z) &= (f \star g)(z) \\
&= \int_{-\infty}^{\infty} f(z-x)\, g(x)\, dx \\
&= \int_{0}^{\infty} f(z-x)\, g(x)\, dx \\
&= \int_{0}^{z} e^{-(z-x)}\, e^{-x}\, dx \\
&= \int_{0}^{z} e^{-z+x}\, e^{-x}\, dx \\
&= \int_{0}^{z} e^{-z}\, dx \\
&= z\, e^{-z} \\
&= \frac{1}{\Gamma(2)\, 1^2}\; z^{2-1}\; e^{-\frac{z}{1}}.
\end{aligned}$$

Hence $Z \sim GAM(1,2)$. Thus, if $X \sim GAM(1,1)$ and $Y \sim GAM(1,1)$, then $X+Y \sim GAM(1,2)$, that $X+Y$ is a gamma with $\alpha = 2$ and $\theta = 1$. Recall that a gamma random variable with $\alpha = 1$ is known as an exponential random variable with parameter θ. Thus, in view of the above example, we see that the sum of two independent exponential random variables is not necessarily an exponential variable.

Example 10.17. What is the probability density of the sum of two independent random variables, each of which is standard normal?

Answer: Let $Z = X + Y$, where $X \sim N(0,1)$ and $Y \sim N(0,1)$. Hence, the density function $f(x)$ of the random variable X is given by

$$f(x) = \frac{1}{\sqrt{2\,\pi}}\, e^{-\frac{x^2}{2}}$$

Similarly, the density function $g(y)$ of Y is given by

$$g(y) = \frac{1}{\sqrt{2\,\pi}}\, e^{-\frac{y^2}{2}}$$

Since X and Y are independent, the density function of Z can be obtained by the method of convolution. Notice that the sum $z = x+y$ is between $-\infty$

and ∞. Hence, the density function of Z is given by

$$\begin{aligned}
h(z) &= (f \star g)\,(z) \\
&= \int_{-\infty}^{\infty} f(z-x)\, g(x)\, dx \\
&= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{\frac{-(z-x)^2}{2}}\, e^{-\frac{x^2}{2}}\, dx \\
&= \frac{1}{2\pi}\, e^{-\frac{z^2}{4}} \int_{-\infty}^{\infty} e^{-\left(x-\frac{z}{2}\right)^2}\, dx \\
&= \frac{1}{2\pi}\, e^{-\frac{z^2}{4}}\, \sqrt{\pi} \left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}}\, e^{-\left(x-\frac{z}{2}\right)^2}\, dx \right] \\
&= \frac{1}{2\pi}\, e^{-\frac{z^2}{4}} \left[\int_{-\infty}^{\infty} \frac{1}{\sqrt{\pi}}\, e^{-w^2}\, dw \right], \qquad \text{where } w = x - \frac{z}{2} \\
&= \frac{1}{\sqrt{4\pi}}\, e^{-\frac{z^2}{4}} \\
&= \frac{1}{\sqrt{4\pi}}\, e^{-\frac{1}{2}\left(\frac{z-0}{\sqrt{2}}\right)^2}.
\end{aligned}$$

The integral in the brackets equals to one, since the integrand is the normal density function with mean $\mu = 0$ and variance $\sigma^2 = \frac{1}{2}$. Hence sum of two standard normal random variables is again a normal random variable with mean zero and variance 2.

Example 10.18. What is the probability density of the sum of two independent random variables, each of which is Cauchy?

Answer: Let $Z = X + Y$, where $X \sim N(0,1)$ and $Y \sim N(0,1)$. Hence, the density function $f(x)$ of the random variable X and Y are is given by

$$f(x) = \frac{1}{\pi\,(1+x^2)} \quad \text{and} \quad g(y) = \frac{1}{\pi\,(1+y^2)},$$

respectively. Since X and Y are independent, the density function of Z can be obtained by the method of convolution. Notice that the sum $z = x + y$ is between $-\infty$ and ∞. Hence, the density function of Z is given by

$$\begin{aligned}
h(z) &= (f \star g)\,(z) \\
&= \int_{-\infty}^{\infty} f(z-x)\, g(x)\, dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\pi\,(1+(z-x)^2)}\; \frac{1}{\pi\,(1+x^2)}\; dx \\
&= \frac{1}{\pi^2} \int_{-\infty}^{\infty} \frac{1}{1+(z-x)^2}\; \frac{1}{1+x^2}\; dx.
\end{aligned}$$

To integrate the above integral, we decompose the integrand using partial fraction decomposition. Hence

$$\frac{1}{1+(z-x)^2}\,\frac{1}{1+x^2} = \frac{2\,A\,x+B}{1+x^2} + \frac{2\,C\,(z-x)+D}{1+(z-x)^2}$$

where

$$A = \frac{1}{z\,(4+z^2)} = C \quad \text{and} \quad B = \frac{1}{4+z^2} = D.$$

Now integration yields

$$\begin{aligned}
&\frac{1}{\pi^2}\int_{-\infty}^{\infty} \frac{1}{1+(z-x)^2}\,\frac{1}{1+x^2}\,dx \\
&= \frac{1}{\pi^2\,z^2\,(4+z^2)}\left[z\ln\left(\frac{1+x^2}{1+(z-x)^2}\right) + z^2\,\tan^{-1}x - z^2\tan^{-1}(z-x)\right]_{-\infty}^{\infty} \\
&= \frac{1}{\pi^2\,z^2\,(4+z^2)}\left[0 + z^2\,\pi + z^2\,\pi\right] \\
&= \frac{2}{\pi\,(4+z^2)}.
\end{aligned}$$

Hence the sum of two independent Cauchy random variables is not a Cauchy random variable.

If $X \sim CAU(0)$ and $Y \sim CAU(0)$, then it can be easily shown using Example 10.18 that the random variable $Z = \frac{X+Y}{2}$ is again Cauchy, that is $Z \sim CAU(0)$. This is a remarkable property of the Cauchy distribution.

So far we have considered the convolution of two continuous independent random variables. However, the concept can be modified to the case when the random variables are discrete.

Let X and Y be two discrete random variables both taking on values that are integers. Let $Z = X + Y$ be the sum of the two random variables. Hence Z takes values on the set of integers. Suppose that $X = n$ where n is some integer. Then $Z = z$ if and only if $Y = z - n$. Thus the events $(Z = z)$ is the union of the pair wise disjoint events $(X = n)$ and $(Y = z - n)$ where n runs over the integers. The cdf $H(z)$ of Z can be obtained as follows:

$$P(Z = z) = \sum_{n=-\infty}^{\infty} P(X = n)\,P(Y = z - n)$$

which is

$$h(z) = \sum_{n=-\infty}^{\infty} f(n)\,g(z - n),$$

where $F(x)$ and $G(y)$ are the cdf of X and Y, respectively.

Definition 10.2. Let X and Y be two independent integer-valued discrete random variables, with pdfs $f(x)$ and $g(y)$ respectively. Then the convolution of $f(x)$ and $g(y)$ is the cdf $h = f \star g$ given by

$$h(m) = \sum_{n=-\infty}^{\infty} f(n)\, g(m-n),$$

for $m = -\infty, ..., -2, -1, 0, 1, 2, ...\infty$. The function $h(z)$ is the pdf of the discrete random variable $Z = X + Y$.

Example 10.19. Let each of the random variable X and Y represents the outcomes of a six-sided die. What is the cumulative density function of the sum of X and Y?

Answer: Since the range of X as well as Y is $\{1, 2, 3, 4, 5, 6\}$, the range of $Z = X + Y$ is $R_Z = \{2, 3, 4, ..., 11, 12\}$. The pdf of Z is given by

$$\begin{aligned}
h(2) &= f(1)\, g(1) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} \\
h(3) &= f(1)\, g(2) + f(2)\, g(1) = \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{2}{36} \\
h(4) &= f(1)\, g(3) + h(2)\, g(2) + f(3)\, g(1) = \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} + \frac{1}{6} \cdot \frac{1}{6} = \frac{3}{36}.
\end{aligned}$$

Continuing in this manner we obtain $h(5) = \frac{4}{36}$, $h(6) = \frac{5}{36}$, $h(7) = \frac{6}{36}$, $h(8) = \frac{5}{36}$, $h(9) = \frac{4}{36}$, $h(10) = \frac{3}{36}$, $h(11) = \frac{2}{36}$, and $h(12) = \frac{1}{36}$. Putting these into one expression we have

$$\begin{aligned}
h(z) &= \sum_{n=1}^{z-1} f(n) g(z-n) \\
&= \frac{6 - |z-7|}{36}, \qquad z = 2, 3, 4, ..., 12.
\end{aligned}$$

It is easy to note that the convolution operation is commutative as well as associative. Using the associativity of the convolution operation one can compute the pdf of the random variable $S_n = X_1 + X_2 + \cdots + X_n$, where $X_1, X_2, ..., X_n$ are random variables each having the same pdf $f(x)$. Then the pdf of S_1 is $f(x)$. Since $S_n = S_{n-1} + X_n$ and the pdf of X_n is $f(x)$, the pdf of S_n can be obtained by induction.

10.5. Moment Generating Function Method

We know that if X and Y are independent random variables, then

$$M_{X+Y}(t) = M_X(t)\, M_Y(t).$$

This result can be used to find the distribution of the sum $X+Y$. Like the convolution method, this method can be used in finding the distribution of $X+Y$ if X and Y are independent random variables. We briefly illustrate the method using the following example.

Example 10.20. Let $X \sim POI(\lambda_1)$ and $Y \sim POI(\lambda_2)$. What is the probability density function of $X+Y$ if X and Y are independent?

Answer: Since, $X \sim POI(\lambda_1)$ and $Y \sim POI(\lambda_2)$, we get

$$M_X(t) = e^{\lambda_1\,(e^t-1)}$$

and

$$M_Y(t) = e^{\lambda_2\,(e^t-1)}.$$

Further, since X and Y are independent, we have

$$\begin{aligned} M_{X+Y}(t) &= M_X(t)\, M_Y(t) \\ &= e^{\lambda_1\,(e^t-1)}\, e^{\lambda_2\,(e^t-1)} \\ &= e^{\lambda_1\,(e^t-1)+\lambda_2\,(e^t-1)} \\ &= e^{(\lambda_1+\lambda_2)(e^t-1)}, \end{aligned}$$

that is, $X+Y \sim POI(\lambda_1+\lambda_2)$. Hence the density function $h(z)$ of $Z = X+Y$ is given by

$$h(z) = \begin{cases} \frac{e^{-(\lambda_1+\lambda_2)}}{z!}\,(\lambda_1+\lambda_2)^z & \text{for } z = 0, 1, 2, 3, ... \\ 0 & \text{otherwise.} \end{cases}$$

Compare this example to Example 10.13. You will see that moment method has a definite advantage over the convolution method. However, if you use the moment method in Example 10.15, then you will have problem identifying the form of the density function of the random variable $X+Y$. Thus, it is difficult to say which method always works. Most of the time we pick a particular method based on the type of problem at hand.

Example 10.21. What is the probability density function of the sum of two independent random variable, each of which is gamma with parameters θ and α?

Answer: Let X and Y be two independent gamma random variables with parameters θ and α, that is $X \sim GAM(\theta, \alpha)$ and $Y \sim GAM(\theta, \alpha)$. From Theorem 6.3, the moment generating functions of X and Y are obtained as $M_X(t) = (1-\theta)^{-\alpha}$ and $M_Y(t) = (1-\theta)^{-\alpha}$, respectively. Since, X and Y are independent, we have

$$\begin{aligned} M_{X+Y}(t) &= M_X(t)\, M_Y(t) \\ &= (1-\theta)^{-\alpha}\; (1-\theta)^{-\alpha} \\ &= (1-\theta)^{-2\alpha}. \end{aligned}$$

Thus $X+Y$ has a moment generating function of a gamma random variable with parameters θ and 2α. Therefore

$$X + Y \sim GAM(\theta, 2\alpha).$$

10.6. Review Exercises

1. Let X be a continuous random variable with density function

$$f(x) = \begin{cases} e^{-2x} + \frac{1}{2}\, e^{-x} & \text{for } 0 < x < \infty \\ \\ 0 & \text{otherwise.} \end{cases}$$

If $Y = e^{-2X}$, then what is the density function of Y where nonzero?

2. Suppose that X is a random variable with density function

$$f(x) = \begin{cases} \frac{3}{8}\, x^2 & \text{for } 0 < x < 2 \\ \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y = mX^2$, where m is a fixed positive number. What is the density function of Y where nonzero?

3. Let X be a continuous random variable with density function

$$f(x) = \begin{cases} 2\, e^{-2x} & \text{for } x > 0 \\ \\ 0 & \text{otherwise} \end{cases}$$

and let $Y = e^{-X}$. What is the density function $g(y)$ of Y where nonzero?

4. What is the probability density of the sum of two independent random variables, each of which is uniformly distributed over the interval $[-2, 2]$?

5. Let X and Y be random variables with joint density function

$$f(x,y) = \begin{cases} e^{-x} & \text{for } 0 < x < \infty;\ 0 < y < 1 \\ \\ 0 & \text{elsewhere .} \end{cases}$$

If $Z = X + 2Y$, then what is the joint density of X and Z where nonzero?

6. Let X be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{2}{x^2} & \text{for } 1 < x < 2 \\ \\ 0 & \text{elsewhere.} \end{cases}$$

If $Y = \sqrt{X}$, then what is the density function of Y for $1 < y < \sqrt{2}$?

7. What is the probability density of the sum of two independent random variables, each of which has the density function given by

$$f(x) = \begin{cases} \frac{10-x}{50} & \text{for } 0 < x < 10 \\ \\ 0 & \text{elsewhere?} \end{cases}$$

8. What is the probability density of the sum of two independent random variables, each of which has the density function given by

$$f(x) = \begin{cases} \frac{a}{x^2} & \text{for } a \leq x < \infty \\ \\ 0 & \text{elsewhere?} \end{cases}$$

9. Roll an unbiased die 3 times. If U denotes the outcome in the first roll, V denotes the outcome in the second roll, and W denotes the outcome of the third roll, what is the distribution of the random variable $Z = \max\{U, V, W\}$?

10. The probability density of V, the velocity of a gas molecule, by Maxwell-Boltzmann law is given by

$$f(v) = \begin{cases} \frac{4\,h^3}{\sqrt{\pi}}\ v^2\ e^{-h^2 v^2} & \text{for } 0 \leq v < \infty \\ \\ 0 & \text{otherwise,} \end{cases}$$

where h is the Plank's constant. If m represents the mass of a gas molecule, then what is the probability density of the kinetic energy $Z = \frac{1}{2}\, mV^2$?

11. If the random variables X and Y have the joint density

$$f(x,y) = \begin{cases} \frac{6}{7}\,x & \text{for } 1 \leq x+y \leq 2,\ x \geq 0,\ y \geq 0 \\ \\ 0 & \text{otherwise,} \end{cases}$$

what is the joint density of $U = 2X + 3Y$ and $V = 4X + Y$?

12. If the random variables X and Y have the joint density

$$f(x,y) = \begin{cases} \frac{6}{7}\,x & \text{for } 1 \leq x+y \leq 2,\ x \geq 0,\ y \geq 0 \\ \\ 0 & \text{otherwise,} \end{cases}$$

what is the density of $\frac{X}{Y}$?

13. Let X and Y have the joint probability density function

$$f(x,y) = \begin{cases} \frac{5}{16}\,xy^2 & \text{for } 0 < x < y < 2 \\ \\ 0 & \text{elsewhere.} \end{cases}$$

What is the joint density function of $U = 3X - 2Y$ and $V = X + 2Y$ where it is nonzero?

14. Let X and Y have the joint probability density function

$$f(x,y) = \begin{cases} 4x & \text{for } 0 < x < \sqrt{y} < 1 \\ \\ 0 & \text{elsewhere.} \end{cases}$$

What is the joint density function of $U = 5X - 2Y$ and $V = 3X + 2Y$ where it is nonzero?

15. Let X and Y have the joint probability density function

$$f(x,y) = \begin{cases} 4x & \text{for } 0 < x < \sqrt{y} < 1 \\ \\ 0 & \text{elsewhere.} \end{cases}$$

What is the density function of $X - Y$?

16. Let X and Y have the joint probability density function

$$f(x,y) = \begin{cases} 4x & \text{for } 0 < x < \sqrt{y} < 1 \\ \\ 0 & \text{elsewhere.} \end{cases}$$

What is the density function of $\frac{X}{Y}$?

17. Let X and Y have the joint probability density function

$$f(x,y) = \begin{cases} 4x & \text{for } 0 < x < \sqrt{y} < 1 \\ \\ 0 & \text{elsewhere.} \end{cases}$$

What is the density function of XY ?

18. Let X and Y have the joint probability density function

$$f(x,y) = \begin{cases} \frac{5}{16}\, xy^2 & \text{for } 0 < x < y < 2 \\ \\ 0 & \text{elsewhere.} \end{cases}$$

What is the density function of $\frac{Y}{X}$?

19. If X an uniform random variable on the interval $[0,2]$ and Y is an uniform random variable on the interval $[0,3]$, then what is the joint probability density function of $X+Y$ if they are independent?

20. What is the probability density function of the sum of two independent random variable, each of which is binomial with parameters n and p?

21. What is the probability density function of the sum of two independent random variable, each of which is exponential with mean θ?

22. What is the probability density function of the average of two independent random variable, each of which is Cauchy with parameter $\theta = 0$?

23. What is the probability density function of the average of two independent random variable, each of which is normal with mean μ and variance σ^2?

24. Both roots of the quadratic equation $x^2 + \alpha x + \beta = 0$ can take all values from -1 to $+1$ with equal probabilities. What are the probability density functions of the coefficients α and β?

25. If A, B, C are independent random variables uniformly distributed on the interval from zero to one, then what is the probability that the quadratic equation $Ax^2 + Bx + C = 0$ has real solutions?

26. The price of a stock on a given trading day changes according to the distribution $f(-1) = \frac{1}{4}$, $f(0) = \frac{1}{2}$, $f(1) = \frac{1}{8}$, and $f(2) = \frac{1}{8}$. Find the distribution for the change in stock price after two (independent) trading days.